

Kyaw (Kelvin) Ha

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EDUCATION

Rutgers University, New Brunswick, NJ
Bachelor of Science, Computer Engineering

Expected May 2027

Ohlone College, Fremont, CA
Associate of Science, Computer Engineering

May 2024

TECHNICAL SKILLS

Languages: Verilog, SystemVerilog, Embedded C, C++, Java, Python, MATLAB, HTML, CSS, JavaScript

Embedded / Firmware: STM32CubeIDE, STM32CubeMX, HAL Drivers, FreeRTOS (CMSIS-RTOS2), UART, PWM, Timer Input Capture, Interrupt-Driven Programming, I2C (in progress)

Digital Design / Tools: FPGA Design (Vivado, in progress), Icarus Verilog, GTKWave, LTSpice, PCB Design, Schematic Capture, Soldering, Logic Analyzers, Oscilloscopes

Core Knowledge: RTL Design & Verification, Microarchitecture, ISA Design, Digital Logic Design, Datapath & Control Design, Real-Time Operating Systems, Circuit Design, PCB Layout & Prototyping, Technical Documentation

PROJECTS

MiniRISC16 Processor Design

Verilog, RTL Design

- Designed a 16-bit RISC CPU from scratch — defined a custom ISA and implemented a single-cycle microarchitecture with a fully custom ALU, register file, instruction decoder, program counter, and instruction memory in RTL.
- Performed architectural tradeoff analysis on datapath and control logic, evaluating design decisions to optimize instruction execution and ensure correct microarchitectural behavior across ADD, SUB, AND, OR, and XOR operations.
- Developed simulation testbenches to verify RTL functionality at the module and integrated level; analyzed VCD waveform traces in GTKWave to debug datapath and control signal behavior throughout the fetch-decode-execute pipeline.

STM32 Embedded Sensor & Control System

STM32CubeIDE, Embedded C, FreeRTOS

- Developed multi-peripheral embedded firmware on an STM32 (Cortex-M0+) microcontroller, integrating GPIO-timed sensor interfacing, hardware PWM generation for servo motor control, and interrupt-driven timer input capture for precision pulse-width signal timing.
- Implemented a FreeRTOS-based multitasking architecture (CMSIS-RTOS2) with independently scheduled concurrent tasks using non-blocking delays, alongside real-time UART telemetry over a serial debug console.
- Diagnosed and resolved HAL driver state-machine and peripheral routing conflicts through systematic register-level debugging, restoring reliable UART communication and timer-interrupt behavior.

Traffic Signal State Machine

Verilog, TTL Logic, Circuit Design

- Implemented a finite state machine in Verilog and realized it as a physical TTL logic circuit on a breadboard — no microcontroller involved, just discrete logic gates driving real LEDs.
- Designed and simulated the circuit schematic in LTSpice and Tinkercad prior to hardware assembly, then built custom clock generation circuitry from discrete logic to drive state register updates.
- Performed functional verification of all state transitions against a working traffic light setup, confirming correct logic behavior across every operating phase.

Huffman Coding Compressor

C++

- Built a lossless file compression and decompression tool using Huffman Trees and priority queues; achieved ~40% size reduction on large text datasets.
- Implemented dynamic encoding for any user-input text file via CLI with full encode/decode round-trip verification, demonstrating strong low-level C++ and algorithmic problem solving.

Personal Portfolio Website

HTML, CSS, JavaScript

- Responsive site with AOS scroll animations, dark-mode toggle, and project detail pages; hosted on GitHub Pages at kelvinthiha17.github.io/portfolio.

EXPERIENCE

R&B Auto Works, Yangon, Myanmar

Jan 2019 – Jan 2020

Technical Intern

- Assisted in automotive diagnostics and hands-on component-level repair, applying systematic troubleshooting to identify and resolve failures.
- Tracked inventory, repair data, and supply usage in Microsoft Excel; documented repair processes and maintained technical accuracy across all maintenance records.

Happy Lemon, Newark, CA

Jan 2023 – Sep 2023

Barista / Team Member

- Managed high-volume orders in a fast-paced environment while maintaining accuracy and customer satisfaction.
- Trained and onboarded new employees on drink preparation, contributing to improved team consistency.